

MAT 426, Class 1: LU Decomposition

Definition.

- A matrix L is **lower triangular** if all entries _____ the main diagonal are _____. In matrix notation, $L = [l_{ij}]$, with $l_{ij} = ______$ for all _____.

$$L = \begin{bmatrix} l_{11} & 0 & 0 & \cdots \\ l_{21} & l_{22} & 0 & \cdots \\ l_{31} & l_{32} & l_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

- A matrix U is **upper triangular** if all entries _____ the main diagonal are _____. In matrix notation, $U = [u_{ij}]$, with $u_{ij} = ______$ for all _____.

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots \\ 0 & u_{22} & u_{23} & \cdots \\ 0 & 0 & u_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Example 1. Determine whether the matrices are upper triangular, lower triangular, neither or both:

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 0 \\ 0 & -2 & 6 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ 8 & 1 & 0 \\ 0 & -1 & 6 \end{bmatrix}$

Using LU Decomposition to Solve Linear Systems

Suppose that $A = LU$ and we want to solve the following system:

• Step 1:

• Step 2:

• Step 3:

Example 2. Use LU decomposition to solve the linear system:

$$\begin{array}{rcrcrcrl} 6x + 18y + 3z & = & 3 \\ 2x + 12y + z & = & 19 \\ 4x + 15y + 3z & = & 0 \end{array}$$

Use the fact that the LU decomposition for the coefficient matrix A of the system is:

$$A = \begin{bmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 6 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 6 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = LU$$

Finding an LU decomposition

Example 3. Let $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$.

- (a) Find the **row echelon form** of the matrix A without using any row interchanges. Check that this matrix is upper triangular. Call it U .

- (b) State the elementary matrices that accomplish this reduction of A to echelon form. You should be able to express U as a product of elementary matrices and A , that is, $U = E_n \cdots E_2 E_1 A$.

- (c) If $U = E_n \cdots E_2 E_1 A$, then $A = E_1^{-1} E_2^{-1} \cdots E_n^{-1} U$. Show that $E_1^{-1} E_2^{-1} \cdots E_n^{-1}$ is lower triangular, call it L . Then we have the desired decomposition of A as LU .

THEOREM *If A is a square matrix that can be reduced to row echelon form without using any row interchanges, then A has an LU factorization.*

An Easier way to find the LU Decomposition

If A can be reduced to row-echelon form _____ row interchanges, then the only row operation operation needed is _____

Note: It is VERY important that we perform the row reduction from _____ to _____ and _____ to _____.

Example 4. Continuing with Example 3...

- Identify the row operations used in Example 3 (in the order applied).
- What do you notice about the multipliers in the row operations and the lower triangular matrix L ?

Example 5. Find the LU factorization of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 8 & 7 & 9 \end{bmatrix}$.

The General Case...

- The product of lower triangular matrices is lower triangular.
- A lower triangular matrix with nonzero diagonal entries is invertible and the inverse is lower triangular.
- Derivation of LU factorization.

ALGORITHM FOR LU FACTORIZATION

- (1) Reduce A to an _____ U by a sequence of _____ operations, if possible.
- (2) Place entries in L such that the _____ sequence of row operations reduce ____ to ____.

Example 6. Find an LU factorization of
$$\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

Example 7. Solve $\begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$ using LU factorization.